

Harsh Gujarathi

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EDUCATION

Georgia Institute of Technology	Atlanta, GA
Master of Science in Computer Science; GPA: 4.0/4.0	August 2026 – May 2028
Current Coursework: Large Language Models, Machine Learning	
Birla Institute of Technology and Science, Pilani	Goa, India
B.E. in Computer Science (Minor in Data Science); CGPA: 9.2/10.0	November 2020 – June 2024

EXPERIENCE

NEXS Lab, Georgia Institute of Technology	Atlanta, GA
Graduate Student Researcher	August 2026 – Present

- Formalizing and evaluating self-speculative decoding for Mixture-of-Experts (MoE) LLM inference, with the current implementation improving output throughput by **1.10–1.21×** across evaluated workloads.
- Built large-scale inference experiments using **vLLM** on **NVIDIA H200 GPUs**, profiling speculative decoding behavior and validating throughput improvements across MoE model configurations.

Microsoft	Hyderabad, India
Software Engineer	July 2024 – July 2026

- Built a RAG-based Azure Migrate assistant with an NL-to-KQL query-generation pipeline, improving agent accuracy from **40% to 85%**.
- Developed Check-In and PreProd evaluation pipelines for the Azure Migrate AI agent, expanding Greenfield (happy-path) coverage from **0% to 100%** and detecting regressions before production deployment.
- Engineered a critical fallback extension for non-container workloads, expanding assessment coverage to **99.9%** across **100K+** enterprise workloads.
- Owned the IIS discovery and assessment microservice end-to-end, combining distributed pub/sub and concurrent task orchestration with versioned REST contracts to support .NET Core, Python, PHP, and Java apps on VMware vCenter.

DevRev	Bengaluru, India
Member of Technical Staff Intern	January 2024 – June 2024

- Engineered Redis-based distributed rate limiting for the Global Notifications microservice (**60K+ daily events**), eliminating drops under bursty load.
- Developed a fault-tolerant deduplication engine using hierarchical clustering, reducing duplicate notification spam by **85%**.
- Designed an LLM-driven summarization pipeline (GPT-4) for digest generation, compressing data throughput by **6x**.

Microsoft	Hyderabad, India
Software Engineer Intern	May 2023 – July 2023

- Instrumented customer-side meter telemetry for the Azure Business Continuity Center using OpenTelemetry and Geneva, enabling real-time observability for V1 services.
- Packaged the instrumentation as a reusable NuGet library with **99% test coverage**, adopted by adjacent teams to accelerate integration.

PROJECTS

Artemis Python, Go, Celery, PostgreSQL, Redis	github.com/artemis-agent
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- Built an autonomous job-discovery agent that detects company ATS platforms and generates scraper configurations, supporting **8** ATS providers across **1,288** companies.
- Designed event-driven ingestion pipelines using Celery, Redis Streams, and pgvector for enrichment, change detection, and semantic job deduplication.

VerbalForge Go, Gin, Python, LangChain, MongoDB, Next.js	www.verbalforge.xyz
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- Built an autonomous assessment-generation agent that creates GRE-style questions, quizzes, and full-length exams across text completion, sentence equivalence, and reading comprehension.
- Designed a Python/LangChain pipeline with schema-based quality validation and stateful async workers, integrated with a Go/Gin REST API, MongoDB, and Next.js.

SKILLS

Languages	Python, C++, Go, C#, SQL
ML & GenAI	Machine Learning, Deep Learning, Natural Language Processing (NLP), PyTorch, TensorFlow, Scikit-learn, LangChain, Large Language Models (LLMs), RAG, LLM Agents, LLM Evaluation, OpenAI, Azure OpenAI
ML Systems	Distributed Systems, vLLM, Model Serving, GPU Inference, Performance Profiling, Speculative Decoding, Mixture-of-Experts (MoE), Vector Search, pgvector
Infrastructure	FastAPI, Docker, Kubernetes, Linux, Azure, AWS, EC2, SageMaker, Bedrock, OpenTelemetry